

Interoperable cloud data platforms for auto-compliant automotive supply chains using digital product passports aligned with Gaia-X and EU ESPR requirements.

Manufacturing × Supply chain risk monitoring × Cloud data platform · OS086 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
56 is the world moving	74 can we play, can we win	HIGH 1 named in the evidence	€2.0bn observed evidence · per year	LATER research and standards activi...	L1 13 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity is about building interoperable cloud data platforms that enable automotive suppliers to automatically comply with EU Digital Product Passport (DPP) regulations, aligned with Gaia-X and ESPR. It exists now because the EU ESPR mandates DPPs, and the market is actively researching how to implement them across complex supply chains.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€2.7bn €490m – €14.6bn	€2.0bn €352m – €10.5bn	€93.8m €16.9m – €504m	observed
Observed: tendered contract value, annualised	€20.9m €20.9m – €56.6m	€20.9m €20.9m – €56.6m	€1.0m €1.0m – €2.7m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (EU27_2020)	238,122 enterprises	observed	Eurostat · sbs_sc_oww	2024
Enterprises adopting Cloud data platform	20.6 % of enterprises (10+ employees)	observed	Eurostat · isoc_cicce_usen2 · E_CC_PDB	2025
Annual value of one engagement	€337k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	39,220 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	4.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

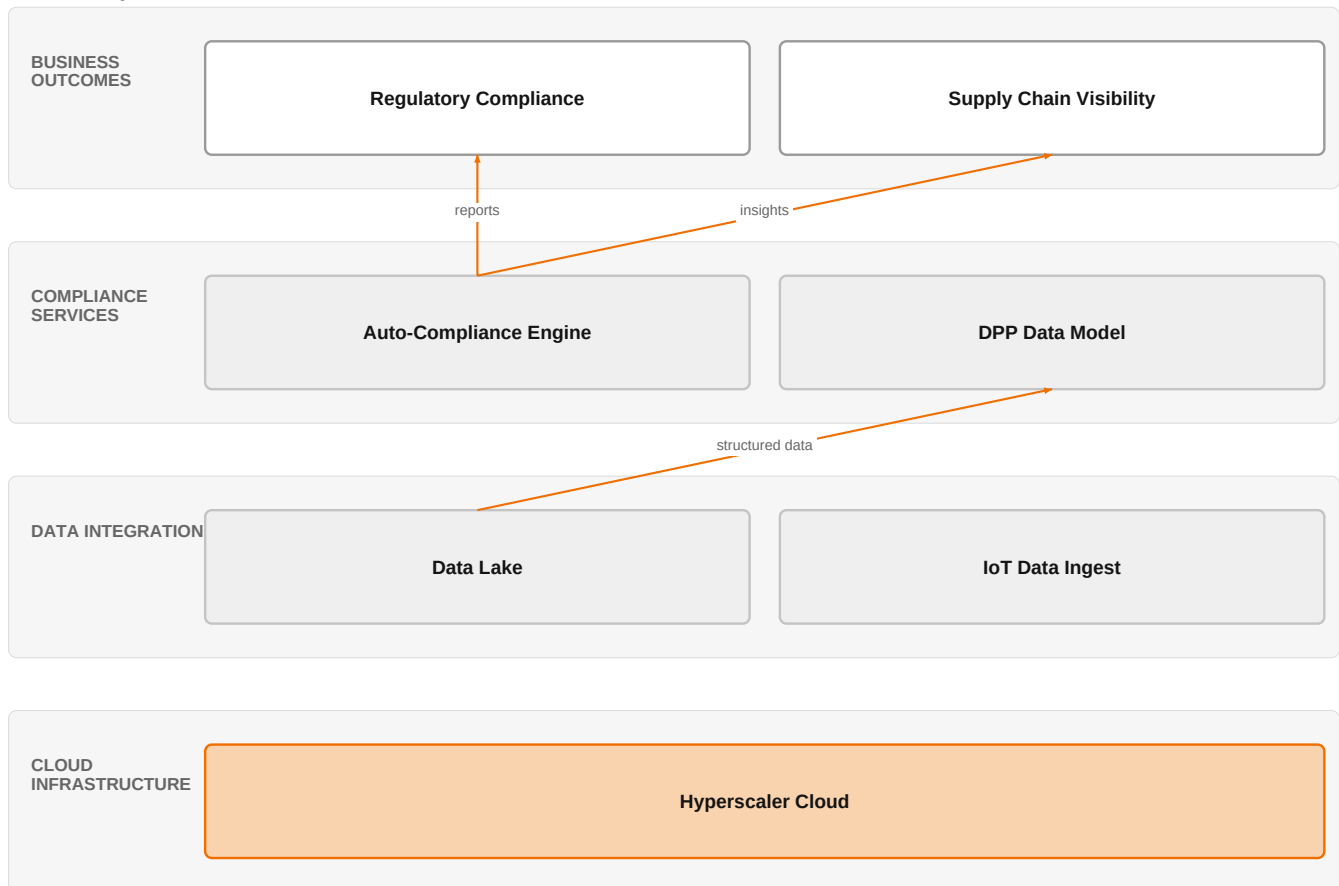
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€20.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	4.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

Auto-Compliant DPP Platform



■ Orange asset
 ■ Partner
 ■ Customer-owned
 ■ Third party / to be sourced

This platform turns raw supply chain data into automated compliance and visibility for DPP regulations.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The EU's Ecodesign for Sustainable Products Regulation (ESPR) is driving the need for Digital Product Passports, which are seen as critical for sustainable supply chains under the European Green Deal. Research shows that DPPs improve traceability, transparency, reliability, and circularity, but full integration is hampered by organizational hurdles. The COMPLIANCE4DPP project is leveraging Gaia-X architectural frameworks to ensure transparency, openness, reliability, security, and interoperability in data exchange with auto-compliance tools along the value chain. This indicates a shift toward standardized, interoperable data ecosystems for regulatory compliance.

Sources: SIG-E4163170FA87 Systems 2026-06-17; SIG-63476A345BB2 cordis.europa.eu 2026-06-01; SIG-89C4D42B6B27 MDPI AG 2025-04-23

Who buys, and why now

The buyer is typically the Chief Digital Officer or Head of Supply Chain at a European automotive manufacturer or tier-1 supplier. The pain is the complexity of collecting and sharing product lifecycle data across a fragmented supply chain to meet ESPR requirements. The trigger is an upcoming regulatory deadline or an audit that reveals non-compliance risks. They need a solution that automates compliance without disrupting existing operations.

Why it is hot now — every claim bound to a dated source

- COMPLIANCE4DPP leverages Gaia-X architectural frameworks to ensure transparency, openness, reliability, security, and interoperability in data exchange with auto-compliance tools along the value chain with DPP. [SIG-63476A345BB2]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a team of AI/Data/Cloud experts and IoT experts to design a cloud data platform that integrates with the customer's existing systems. The platform would use Orange's ISO 27001 and TISAX certifications to ensure security and compliance. Orange would leverage its partnerships with Microsoft, AWS, and Google Cloud to provide the underlying cloud infrastructure. The engagement would start with a discovery phase, then build a data model for DPPs, integrate with suppliers' systems, and finally deploy auto-compliance tools.

Why Orange can win

Orange can win because it brings certified security (ISO 27001, TISAX) and deep expertise in AI, data, and cloud, combined with IoT capabilities. Its partnerships with major hyperscalers allow flexibility in cloud choice. However, the assets are thin on specific DPP domain expertise; Orange would need to partner or build that knowledge.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Orange Cyberdefense managed services	offer	L1	offer addresses the use case but not this technology	unconfirmed
Microsoft	partner	L2	partner provides the required technology	unconfirmed
AWS	partner	L2	partner provides the required technology	unconfirmed
Google Cloud	partner	L2	partner provides the required technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Healthcare experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed
Saint-Gobain Glass	reference	SUP	published reference in this vertical, different use case	unconfirmed
Skytale	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

HIGH competitive intensity (score 7.8; 1 of 8 listed players appear in this topic's own evidence)

Crowded, with heavyweight incumbents already visible in the evidence. Win on a specific differentiator or do not bid.

Competitors include hyperscalers like AWS, Google Cloud, and Microsoft, who offer cloud data platforms but may lack vertical-specific DPP solutions. Global system integrators like Accenture, Capgemini, DXC Technology, and Infosys are also active in manufacturing and cloud platforms. Huawei is noted in the evidence for smart manufacturing with 5G and AI, though it is also an Orange partner. The customer will compare based on existing cloud relationships, industry expertise, and ability to handle regulatory complexity.

Sources: SIG-2FF6CAD27F35 developingtelecoms.com 2026-08-06

Competitor	Category	Position here	Basis	Evidence
Huawei	Network equipment vendor	named in this topic's evidence — also an Orange partner	evidenced	SIG-2FF6CAD27F35
AWS	Hyperscaler	sells Cloud data platform; competes in Cloud, OX: Smart Industries — also an Orange partner	structural	—

Google Cloud	Hyperscaler	sells Cloud data platform; competes in Cloud — also an Orange partner	structural	—
Microsoft	Hyperscaler	sells Cloud data platform; competes in Cloud — also an Orange partner	structural	—
Accenture	Global systems integrator	sells Cloud data platform; active in Manufacturing; competes in Cloud, OX: Smart Industries	structural	—
Capgemini	Global systems integrator	sells Cloud data platform; active in Manufacturing; competes in Cloud, OX: Smart Industries	structural	—
DXC Technology	Global systems integrator	sells Cloud data platform; competes in Cloud	structural	—
Infosys	Global systems integrator	sells Cloud data platform; competes in Cloud	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What is your current readiness for ESPR compliance, and what is your target date for DPP implementation?
- 2 How many tiers of suppliers do you need to collect data from, and what is the current state of their data systems?
- 3 Have you already selected a cloud provider for your data platform, and if so, which one?
- 4 What is your biggest pain point today: data collection, data standardization, or compliance reporting?
- 5 Are you participating in any industry pilots or working groups on DPP standards?
- 6 Who in your organization would own the DPP initiative, and what budget is allocated?

Objections you will meet

They will say	You can answer
We already have a cloud data platform from a hyperscaler; why do we need Orange?	That's true, and we work with those hyperscalers as partners. What we add is the integration layer, the compliance expertise, and the certifications (ISO 27001, TISAX) that make the platform ready for regulatory requirements.
DPP is still not a clear regulatory requirement for us yet.	You're right that the final rules are still being defined, but the direction is clear. Starting now with a flexible architecture that can adapt to evolving standards will save you from costly rework later.
Our suppliers won't share data with us.	That's a common challenge. Our approach includes data governance and interoperability frameworks that make it easier for suppliers to share only what's needed, while maintaining their own control.

Risks and unknowns

The main risk is that DPP standards are still evolving; the customer may not yet have a clear regulatory requirement. There is also uncertainty about data sharing across multiple tiers of the supply chain, and whether suppliers will cooperate. Orange's lack of a proven DPP reference could stall deals.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief that differentiates Orange's offering by combining Microsoft, AWS, and Google Cloud interoperability with Gaia-X frameworks, using the Colgate-Palmolive and Saint-Gobain Glass references to demonstrate auto-compliance in supply chain data exchange.
Sales	In a conversation with a tier-1 automotive supplier's head of supply chain, say: 'Orange Cyberdefense managed services can help you meet EU ESPR requirements by securing your digital product passport data exchange, leveraging our experience with Colgate-Palmolive and Saint-Gobain Glass.'
Strategist / Innovator	Commission a deep-dive brief on how Orange's AI/Data/Cloud and IoT experts can extend the Colgate-Palmolive and Saint-Gobain Glass supply-chain data platforms into a Gaia-X-aligned digital product passport offering for automotive, validating against ISO 27001 and TISAX certifications.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-E4163170FA87	2026-06-17	Systems	1	Unlocking Digital Product Passport Integration: Multidimensional Hurdles in Supply Chains
SIG-63476A345BB2	2026-06-01	cordis.europa.eu	1	Fostering Resilient and Auto-Compliant Manufacturing Value Chains through Advanced Digital Prod...
SIG-36FFB2435F8A	2026-04-20	openalex.org	1	A Comprehensive Review of Digital Product Passports in Sustainable Manufacturing: Current State...
SIG-F47C31AB60CB	2026-02-14	MDPI AG	1	Dynamic Traceability and Reliability Management: Integrating the Digital Product Passport with ...
SIG-4FE21FD2ED06	2026-01-01	openalex.org	1	On the Utilization of Digital Product Passports for Intelligent Metal Sorting and Recycling
SIG-CC28A63D0BB7	2026-01-01	openalex.org	1	From Compliance to Circularity: Stakeholder Perspectives on Adopting Digital Product Passports ...
SIG-8123897AE6E6	2025-12-15	openalex.org	1	Bridging digital tools and human skills: A sustainable competence framework for implementing di...
SIG-2B47C056B7E5	2025-10-30	openalex.org	1	Digital material passports in the AEC industry: a scoping review and conceptualisation of futur...
SIG-9BD5C9787E89	2025-10-30	openalex.org	1	Research on a Robust Traceability Method for the Assembly Manufacturing Supply Chain Based on B...
SIG-40DC35D6534F	2025-10-14	openalex.org	1	Predictive Generation of Digital Product Passport as Decision Support in Circular Networks
SIG-6FBECA0F7EA7	2025-09-10	openalex.org	1	Design of a Sensor-Based Digital Product Passport for Low-Tech Manufacturing: Traceability and ...
SIG-89C4D42B6B27	2025-04-23	MDPI AG	1	Digital Product Passport—EU Sustainability and Circularity Regulation
SIG-2FF6CAD27F35	2026-08-06	developingtelecoms.com	2	AIS and Huawei Cloud team for smart manufacturing powered by 5G and AI

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (3 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:14+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:17+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:56:54+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.